

Andrew Aburustum

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EDUCATION

Franklin W. Olin College of Engineering <i>B.S. in Electrical and Computer Engineering (intended)</i>	Needham, MA <i>Expected May 2030</i>
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EXPERIENCE

Memorial Sloan Kettering Cancer Center <i>Biomedical Engineering Intern, formal rotation under Paul Booth</i>	New York, NY <i>Summer 2025</i>
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- Developed post-processing techniques for biomedical implants and machined parts for medical devices
- Operated Stratasys and resin 3D printers and helped repair a plasma cutter

Olin College IT Department <i>Student IT Technician</i>	Needham, MA <i>Fall 2026 - Present</i>
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- Troubleshoot and repair student and staff devices at the campus IT help desk

ENGINEERING TEAMS

Olin Baja <i>Electrical Team Member</i>	Needham, MA <i>Fall 2026 - Present</i>
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- Co-developing a non-contact fuel-level estimator using accelerometers, a custom PCB, and a Python model

FIRST Tech Challenge Team 14450 RoboRebels <i>Team Captain</i>	Saddle River, NJ <i>2022 - 2026</i>
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- Led the team to win the international 2025 Mexico Premier Event as the 1st seed (6-0)
- Mentored FTC teams across the world, including teams in Australia, Morocco, and Venezuela
- Designed, built, and programmed 3 robots with linear slide, arm, and flywheel auto-targeting systems
- Two-time NJ State Championship finalist; won 2 Innovate and 2 Control Awards

PROJECTS

AI & Machine Learning Projects <i>Python, PyTorch, TensorFlow, Keras</i>	github.com/mumtez/AIfeb14
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- Trained VAE, GAN, PixelCNN, and diffusion models in PyTorch to generate weather and Pokémon images
- Developed dictionary-based, statistical, and neural network sentiment classifiers
- Built weather and facial image classifiers

Homelab | *Proxmox, Docker, Portainer, Nginx Proxy Manager*

- Build Docker images and Compose stacks with container networks and health checks
- Run services on a Proxmox host through Portainer, routed over HTTPS by Nginx Proxy Manager

FTC Event Viewer <i>TypeScript, Docker, FTCScout API</i>	ftc.aburustum.com
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- Built a scouting web app that sorts the teams at any FTC event by offensive power rating (OPR)

Console & GPU Repair | *BGA rework, SMD soldering*

- Repaired game consoles and GPUs, replacing HDMI ports and GPU memory chips

SKILLS

Software:	Python, Java, TypeScript, MATLAB, PyTorch, TensorFlow, Keras, Git, Docker
CAD & PCB Design:	Fusion 360, Onshape, SolidWorks, KiCad, Altium Designer
Fabrication:	3D printing (FDM, resin, Stratasys), machining, SMD soldering, BGA rework, reflow
Languages:	Italian (intermediate, B1), Arabic (elementary, A2)

ACTIVITIES & VOLUNTEERING

Olin Handball , Division 2 Athlete	Fall 2026 - Present
EMS Summer String Festival , Volunteer Camp Counselor	Summers 2023 - Present
Community Chest of Eastern Bergen County , Volunteer	2018 - Present